QP CODE: 25901161

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course(DSC)–MG4DSCCND201- LIFE CYCLE AND WELLNESS NUTRITION-II

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S),Interest (I) and Appreciation(Ap)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer all questions
Each question carries **1** mark

1. b). Unhealthy eating habits
2. c). All nutrients in adequate amounts
3. a). Role of metabolism in ageing
4. b). 43 g
5. b). Hypothyroidism
6. b). Sarcopenia

Part B
Short Answer Type Questions.
Answer any **2** questions.
Each question carries **2** marks.

7. Discuss why Junk foods should be avoided in Adolescent diet [U]/[CO1]

- Junk foods are rich in calories, fats,sugar and salt but very low in essential nutrients like fibre,vitamins and minerals
- Frequent intake of junk foods can lead to obesity,hypertension,high cholesterol and even early onset of diabetes
- It may also cause digestive issues,acne and poor dental health.**(Answer any 2 points,1 mark for each point)**

8. Define the term Indian Reference Man as per ICMR [K]/[CO2]

Indian Reference man is between 19-39 years of age and weighs 65 kg with a height of 1.77m with a BMI of 20.75 and is free from disease and physically fit for active work,on each working day,he is engaged in 8 hours of occupation which usually involves moderate activity,while when not at work he spends 8 hours in bed,4-6 hours in sitting and moving about,2 hours in walking and in active recreation or in household duties**.(2 marks)**

9. List two benefits of adequate protein intake in the elderly. [K]/[CO2]

- Adequate protein intake supports wound healing-1 mark
- Immune function- 1 mark
- Muscle strength, and mobility- 1 Mark **(Any 2 points)**.

10.State why breakfast is called the "most important meal of the day"[K]/[CO1]

- It is the first meal eaten after a night of fasting and provides the body with essential nutrients and energy to start the day.
- For adolescents,who are undergoing rapid physical,mental and emotional growth,breakfast plays a crucial role in supporting overall health,academic performance and development.**(I mark for each point)**

Part C

Short Essay Type Questions.
Answer any**4** questions.
Each question carries **5** marks.

11. Explain the developmental features of Middle Adolescence [U]/[CO1]

- **Physical Growth:**
  - Most pubertal changes are completed in this stage
  - Physical appearance becomes more adult like
  - Growth rate slows down by the end of this stage
  - Boys continue to develop muscles and strength,girls gain feminine body shape

- **Cognitive Development:**
  - Abstract and logical thinking develops
  - Can think about the future and make simple plans
  - Begin to form personal values,beliefs and moral ideas
  - Can think about right and wrong but may act impulsively

- **Emotional and Social Development**
  - Strong desire for independence and privacy
  - Peer influence is strongest in this stage
  - Friendships and romantic relationships become very important
  - May show risk taking behaviour and defiance
  - Feel self-conscious about appearance and body image **(Any 5 points,1 mark for each point)**

12.Explain major causes of Diabetes Mellitus in Adults. [U]/[CO2]

1. **Genetic Factors**
   - Family history of diabetes increases the risk
   - Certain ethnic groups (eg: South Indians)have a higher genetic predisposition **(1/2 mark for each point)**

2. **Lifestyle Factors**
   - Unhealthy diet
   - Physical inactivity
   - Obesity:Particularly abdominal obesity is a major risk factor for insulin resistance**(1 mark for any 2 points)**

3. **Physiological and Medical Conditions**
   - Age:Risk increases after 40 years
   - Hypertension and Dyslipidemia
   - Gestational diabetes
   - Polycystic Ovarian Syndrome**(1/2 mark for each point)**

4. **Environmental and Psychological Factors**
   - Stress: Chronic stress can raise cortisol levels,affecting glucose metabolism
   - Sleep deprivation:Alters glucose tolerance and appetite regulation.**(1/2 mark for each point)**

13. Discuss the dietary modifications required to prevent Sarcopenia in old age. [U]/[CO3]

- Include high quality protein in the diet-milk, egg, fish, pulses and lean meat- helps to maintain muscle mass

- Even distribution of protein in all meals (Breakfast, lunch, dinner)- improves muscle protein synthesis.
- Ensure sufficient vitamin D and Calcium-Sunlight exposure, Milk and milk products-supports muscle function.
- Anti-oxidant rich foods – Fruits and vegetables-reduce oxidative stress that contribute to muscle loss.
- Adequate energy intake – Healthy fats & complex carbohydrates-sufficient calories to prevent muscle break down.

14.Discuss the nutritional risks associated with teenage pregnancy.[U]/[CO1]

- Child mothers are considered 'high risk'as maternal and perinatal death rates and infant mortality rates are higher among them.
- Nutritional deficiency signs get aggravated if the adolescent girl conceives immediately after menarche
- She has to meet the demands of not only her growth but also that of conceptus
- Both preclampsia and eclampsia are found to be higher and so also the incidence of pregnancy wastages
- Lactation failure
- Due to small size of the pelvis- increase in the rate of premature deliveries and high incidence of low birth weight**.(1 mark for any 5 points)**

15. Discuss any five dietary guidelines to reduce the cost of a meal [U]/[CO2]

- Cereals,which are less expensive can be increased to more than the normal amount
- Inclusion of millets like ragi,jowar and bajra
- Big and thick rotis can be prepared to save time and fuel
- Combination of cereals and pulses improves the quality of cereal as well as pulse protein
- Pulses like horse gram and cow-pea can be included
- Fermenting,malting and sprouting can enhance the nutritive value
- Greens,particularly from trees like drumstick or agathi are cheaper
- Locally available or kitchen garden produce can be used.
- Inexpensive and nutritious fruits like pappaya and guava can be included
- Food stuffs distributed through public distribution system(Ration shops)can be used **(Any 5 points,1 mark for each point)**

16. Discuss the role of cereals,vegetables and fruits in adult diet [U][CO2]

- **Cereals**
  - Energy derived from cereals should not be more than 75 percent
  - It is better to include two cereals in one meal like rice and wheat or millets and rice
  - Whole grain cereals,brown rice,hand pounded rice,unpolished or parboiled rice or malted grains have high nutritive value and higher phytochemical value.
  - Refined cereals should be taken in minimum quantity.**(1/2 mark for each point)**

- **Vegetables and fruits**

  - Inclusion of salads and raita not only help meet the vitamin requirements but the meals would be attractive and high satiety value due to the fibre content
  - Green Leafy Vegetables should be taken minimum of one serving per day
  - Five servings of colourful fruits and vegetables should be included in a day's diet to meet antioxidant requirement
  - Fruits should never be cooked
  - It is better to serve the fruit as such.Taking juice out of it causes loss of Vitamin C and fibre.
  - Every meal should contain atleast one medium size fruit.**(1/2 mark for each point)**

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Enumerate the health problems associated with poor eating habits in adolescents [K]/[CO1]

- **Skipping Meals**

- Skipping meals deprives the body of essential nutrients such as proteins,iron,calcium and vitamins that are needed for growth,energy and concentration
- As a result,it can cause fatigue,poor academic performance and reduced alertness
- It also leads to overeating later in the day,increasing the risk of obesity and digestive problems.

- **Excessive consumption of Junk foods**

- Junk foods are very low in essential nutrients like fibre,vitamins and minerals.
- Frequent intake of junk food can lead to obesity,hypertension,high cholesterol and even early onset of diabetes

- **Consumption of soft drinks and sugary beverages**

- These drinks are loaded with sugar and caffeine,which contribute to obesity,tooth decay and disturbed sleep patterns.
- They can also cause metabolic disorders and weaken bones because of decreased calcium intake.

- **Consumption of foods containing trans fats and processed ingredients**

- Trans fats increase bad cholesterol (LDL) and reduce good cholesterol(HDL),leading to obesity,heart disease and fatty liver.
- They also trigger inflammation,hormonal imbalance and fatigue ,negatively affecting both physical and mental health.

- **Fad diets**
- Prolonged adherence to such diets can cause nutritional deficiencies,weakness,hormonal disturbances and even eating disorders such as anorexia and bulimia.
- These diets can interfere with normal growth and physical development during adolescence.

- **Alcohol Consumption**
- It affects brain development,impairs judgement and increases the risk of accidents
- Long term alcohol consumption can damage the liver,reduce appetite and interfere with nutrient absorption ,especially B-vitamins and folic acid.**(Answer any 10 points,1 mark for each point)**

18. Describe the nutritional recommendations during Adulthood [K][CO2]

- **Energy:**

- Caloric needs increase with the metabolic demands of growth and energy expenditure.
- Boys need 2860-3320kcal a day and girls need 2400-2500 kcal a day.

- **Proteins:**
- Boys need 45-55gm/day while girls need 43-46gm/day
- This meets growth needs and for the pubertal changes in both sexes and for the developing muscle mass in boys

- **Fat and Essential Fatty acids:**
- The desirable level of visible fat intake for adolescents is 35-50g/d
- The essential fatty acid requirements is 3% E for adolescents.

- **Minerals:**
- During adolescence,1000-1050mg of calcium and 1000-1050 mg of phosphorus per day is needed.Bone growth demands calcium and phosphorus
- Boys require 22-26 mg/day and girls require 30-32mg/day of iron
- Zinc requirement is 14.3 to 17.6mg/day for boys and 12.8 to 14.2 mg/day for girls.

- **Vitamins:**

- A daily allowance of 860-1000microgram of retinol is recommended for adolescents.
- 600IU of Vitamin D is recommended for adolescents.

- Folacin and Vitamin B12 are essential for DNA synthesis .For all adolescents 2.2 microgram of Vitamin B12 is recommended**.(Answer any 10 points,1 mark for each point)**

19. Explain the causes and health consequences of obesity in elderly.

Causes (Any 5 points. 1 Marks for each point)

- Reduced Basal Metabolic Rate
- Reduced physical activity
- Over consumption of energy dense foods
- Hormonal changes
- Medications
- Emotional/psychological issues
- Social factors
- Genetic predisposition
- Chronic diseases

Health Consequences (Any 5 points. 1 Marks for each point)

- Cardiovascular diseases
- Type II Diabetes Mellitus
- Dyslipidaemia
- Osteoarthritis
- Reduced mobility
- Increased risk of cancers

20. Describe the causes and types of malnutrition in elderly with suitable examples. [U][CO3]

Causes

- Physiological factors
  - Reduced appetite
  - Reduced sense of taste and smell
  - Dental problems
  - Gastro intestinal changes
  - Chronic illness **(Any 3 points. 0.5 mark for each point)**
- Psychological factors
  - depression
  - Cognitive decline/dementia
  - Anxiety and stress **(0.5 mark for each point)**
- Socio economic factors
  - Social isolation
  - Low income
  - Lack of care giver support **(0.5 mark for each point)**
- Functional limitations
  - Reduced mobility
  - Physical disabilities
  - Swallowing difficulties/Dysphagia **(0.5 mark for each point)**
- Medication related factors
  - Drugs affecting appetite
  - Drugs causing nutrient malabsorption and poly pharmacy **(0.5 mark for each point)**

Types (1 mark for each point)

- Undernutrition
- Over nutrition
- Micronutrient deficiencies